ENGINEERING MECHANICS

Instructor: Tongtong GUO

College of Science

Background and Schedule

- Course Name: Engineering Mechanics
- Prerequisites: Advanced Mathematics, Mechanics in College Physics
- Subsequent Courses: Elastic Mechanics, Solid Mechanics, Computational Mechanics, etc.
- Graduate Courses: Continuum Mechanics, Nonlinear Finite Element Method, etc.
- Class Hours: 48 hours
- Instructors: Tongtong GUO (32 hours), Baolin WANG (16 hours)
- Email:tongtong.guo@hit.edu.cn
- Phone: 13926501601 (WeChat)

Background and Schedule

Class Time: 12 Weeks (Spring Semester 2026) 4 hours / week

Midterm Exam: 9 - 10th week, Final Exam: Around Week 15 - 16 (Academic Affairs Office Notice)

Grading: Regular Performance (including quizzes, midterm, attendance, and assignments) accounts for 40%, Final Exam accounts for 60%.

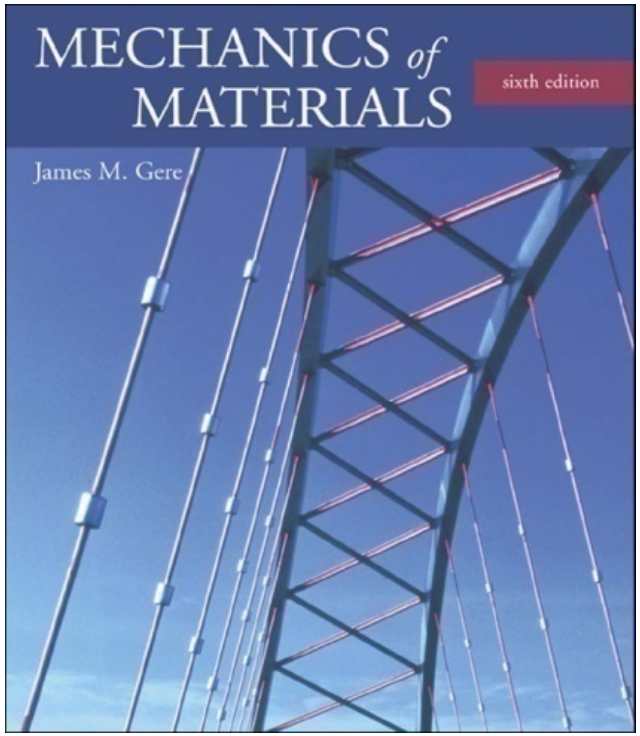
Course Content: Statics, Mechanics of Materials

Reference Books

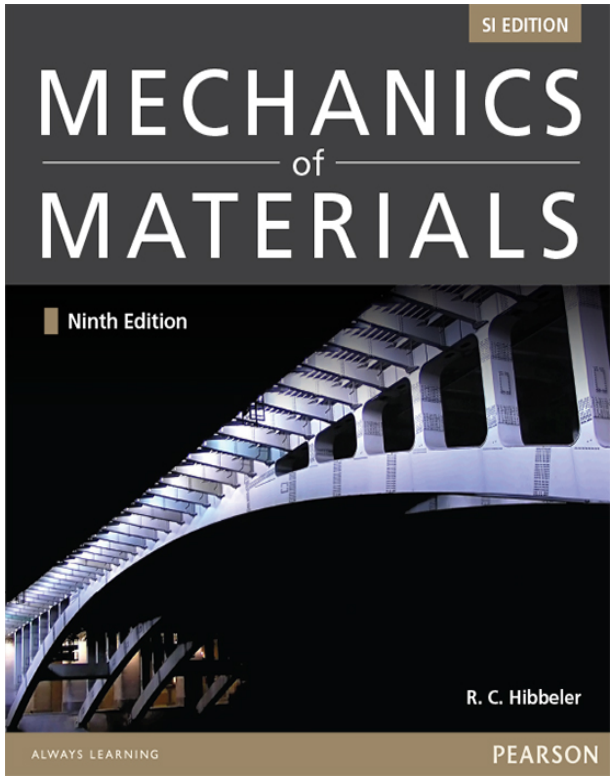
- Title: **Engineering Mechanics STATICS, 15th edition**

Author: R.C.HIBBELER

Background and Schedule



Authors: James M. Gere
Title: Mechanics of Materials
Edition: 6th SI Edition



Authors: RC Hibbeler
Title: Mechanics of Materials
Edition: 9th SI Edition

Background and Schedule

Class rules :

No mobile phones in class, must be turned off or muted

Must attend lectures (apply for leave only for valid reasons).

Excessive absences (more than 2-3 consecutive times) will result in consequences for the student!

(ask for your college counselor sign on the written request for leave)

No lateness or early leave! No reading other materials, eating, or sleeping in class!

Keep the classroom quiet.

No plagiarism allowed.

§0. Introduction

- **Mechanics and Its Development**
- **Scope and Content of Theoretical Mechanics**
- **Content of Statics**

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- **Mechanics and Its Development**
  - **Force and Mechanics**
  - **A Brief History and Branches of Mechanics**
  - **Development Trends of Mechanics**
  - **Applications of Mechanics in Engineering and Life**
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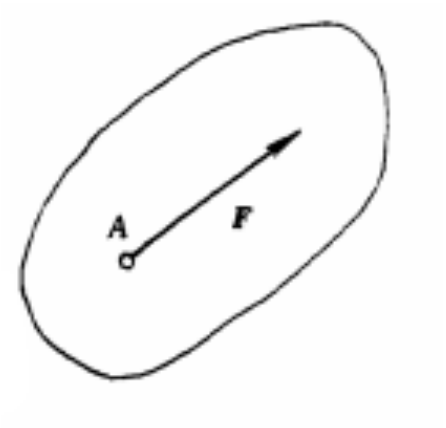
§0. Introduction

- **Force -** a mechanical interaction between objects
- The three **elements** of **force** —

  magnitude, direction, and point of application



Force is a positioning vector with a fixed point of action.

Force：represented as **a vector**

The force vector is represented by a directed line segment:

The length of the line segment represents the magnitude of the force;

The direction of the line segment and the arrow represent the direction of force;

The starting or ending point of a line segment represents the point of application of force.

The black italic ***F*** represents the force vector,

The regular font **F** represents the magnitude of force.

§0. Introduction

- **Mechanics -** the science of studying the effects of force on the motion or deformation of objects.

- Mechanics is the science of force, motion, and deformation, which studies the macro/micro mechanical behavior of complex media in nature and engineering, revealing the laws of mechanical motion and its interactions with physical, chemical, and biological processes. It is an important component of the human scientific knowledge system.

- Mechanics reveals the deformation laws, motion forms, and interaction laws of matter, providing theories and methods for human understanding of natural phenomena, life phenomena, engineering analysis, and design. It plays an important leading, supporting, and promoting role in the development of engineering science.

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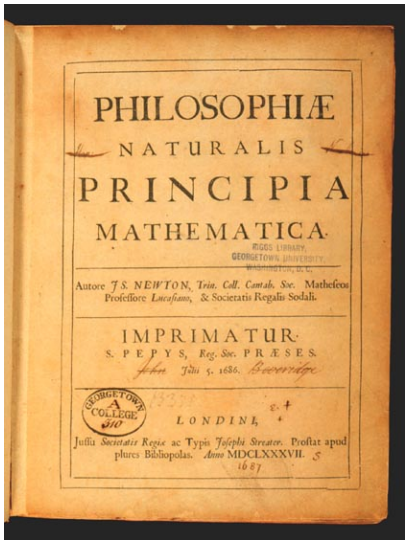
A Brief History of the Development of Mechanics

1. As early as (287-212 BC) in ancient Greece, Archimedes' "On Gravity" laid the foundation for statics.
2. Leonardo da Vinci (1452-1519) from Italy studied sliding friction, balance, and torque.
3. Copernicus of Poland (1473-1543) established the heliocentric theory of the universe - Human tentacles are not limited to the Earth itself. （opposite geocentric）
4. Kepler (1571-1630) from Germany proposed three laws of planetary motion.
5. Galileo of Italy (1564-1642) introduced the concepts of free fall law, inertia law, and acceleration.

§0. Introduction

A Brief History of the Development of Mechanics

6. The great British scientist Newton (1643-1727) summarized his work in the 1687 edition of "Mathematical Principles of Natural Philosophy" by proposing three fundamental laws of motion, the law of universal gravitation and celestial mechanics. He is the founder of classical mechanics.



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A Brief History of the Development of Mechanics

The progress of mechanics thereafter lies in the transformation of the objects it considers from individual free particles to constrained particles and systems of constrained particles; The hallmark of this aspect is the D'Alembert principle proposed by J.R. D'Alembert and the analytical mechanics established by J.L. Lagrange.

- **7. Bernoulli of Switzerland (1667-1748) proposed the principle of virtual displacement.**
- **8. Euler of Switzerland (1707-1783) wrote "Mechanics" to study differential equations.**
- **9. D'Alembert's (1717-1785) classic work "On Dynamics" is the D'Alembert Principle.**
- **10. Lagrange (1736-1813) proposed the second type of Lagrange equation in France.**

§0. Introduction

The three branches of mechanics:

Finite Degree of Freedom Problem (Particle System/Rigid Body Mechanics) — General Mechanics

Rigid body motion, analytical mechanics, dynamical systems, and motion stability.
(Euler, Poincaré, Lagrange, Hamilton, etc.)

Continuum Mechanics — Solid Mechanics
(Euler, Gerhoff, Hooke, Poisson, Cauchy, Saint Venant, etc.)

Continuum Mechanics — Fluid Mechanics
(Bernoulli, Euler, Renault, Raleigh, etc.)

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The Development and Branches of Mechanics:

1. Based on experimental observation
2. Experimental observation must be elevated to theory (mechanical model)
3. Theory must be described in mathematical terms (mathematical models)
4. Quantitative results must be obtained through mathematical solutions.

- **— Research ideas and methods of mechanics（all science）**
- **— Mechanics is a science, not a technology, and certainly not just experience**

§0. Introduction

Modern mechanics in the 20th century:

Challenges to classical mechanics: Ethereum drift, UV catastrophe

- Relativity Mechanics (Einstein)
- Quantum mechanics (Planck, Einstein, De Broglie, Schr ö dinger, Heisenberg, Dirac)

The second milestone in the history of mechanics!

Mechanics is separated from physics:

Physics focuses more on the natural laws of the **microscopic** world
Mechanics focuses more on the natural laws of the **macroscopic** world

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The integration of mechanics and mathematics is tighter, driven by industrial development, permeating into other natural and engineering sciences, and the branch disciplines are more diverse：

Theoretical mechanics, material mechanics, fluid mechanics,

Elasticity, viscoelasticity, plasticity, fracture mechanics, damage mechanics, vibration mechanics, wave dynamics, composite material mechanics, micromechanics, explosion mechanics, structural mechanics, geotechnical mechanics, plasma mechanics, physical mechanics, aerodynamics, hydrodynamics, seepage mechanics, physical chemical fluid mechanics, biomechanics …

Several major achievements in mechanics of the 20th century:

- Boundary layer theory
- Nonlinear mechanics (chaos, bifurcation, solitary waves)
- Fracture mechanics
- Computational mechanics (finite element)

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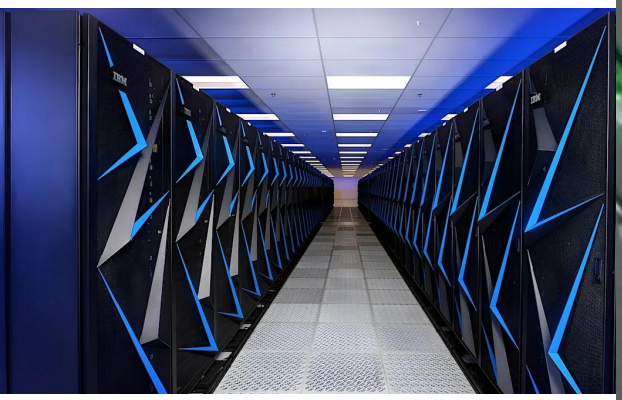
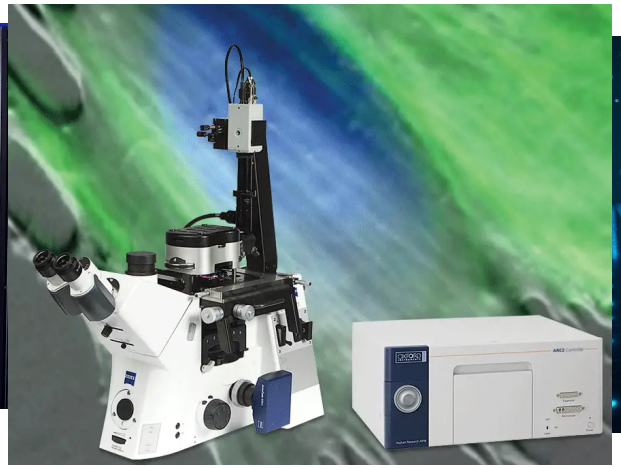
The development of mechanics - the trend of mechanics development in the 21st century.
There are still two unresolved problems: turbulence and material strength

Innovation in research methods:

The Application of Modern Mathematics

- Improvement of computing power (development of computer technology)
- Improvement of experimental observation ability (development of high-tech)





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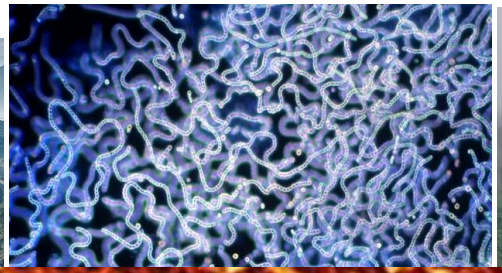
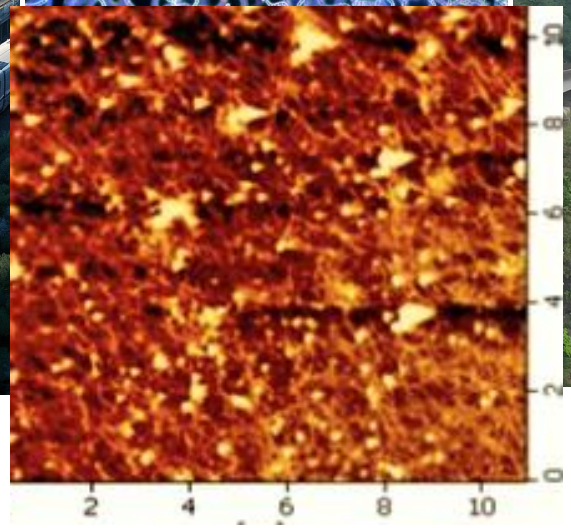
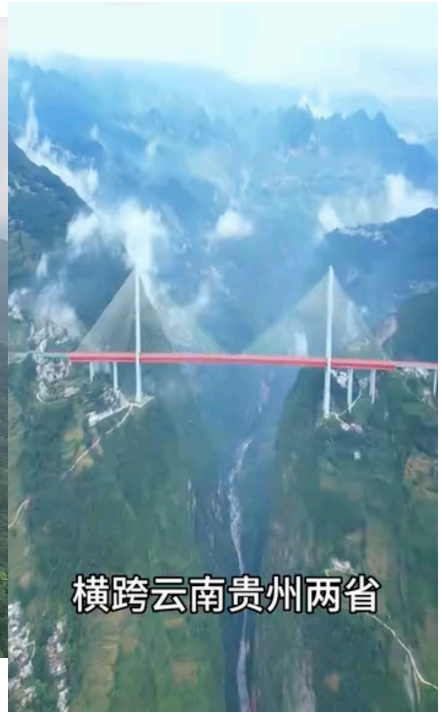
Further expansion of research content

- (1) It no longer stays at the macro level, emphasizing the micro, meso, nano, cross scale correlation, and multi-scale analysis

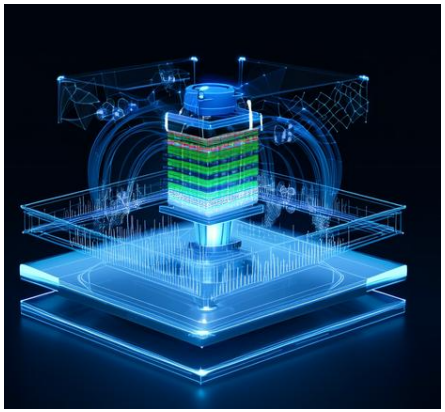


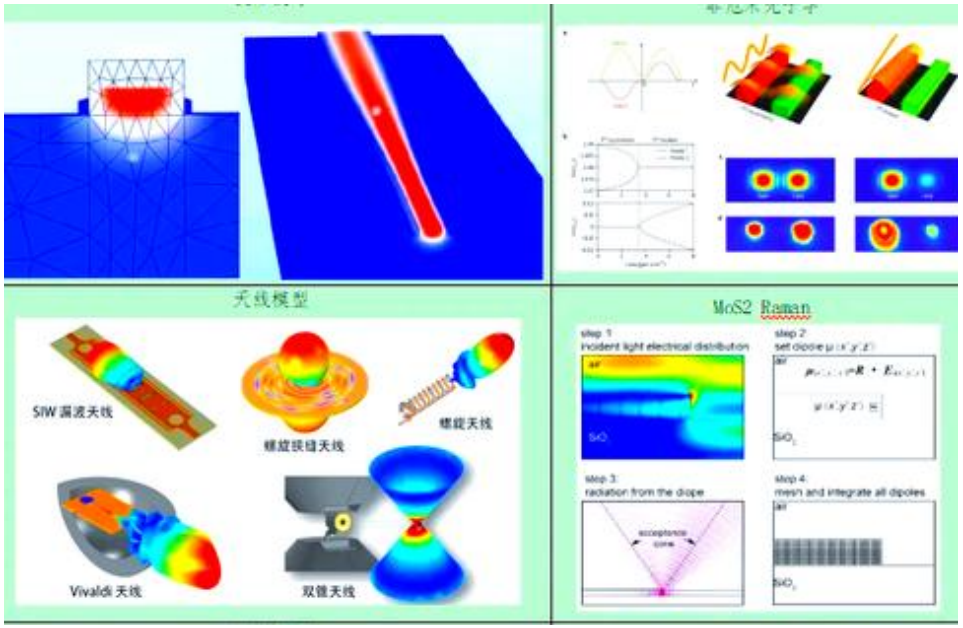


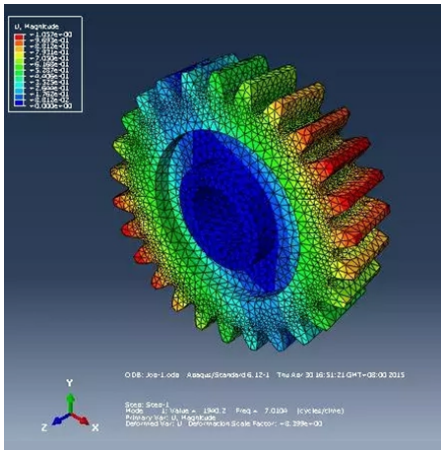


§0. Introduction

- (2) Multi physics field coupling, including electrical, magnetic, acoustic, thermal, and chemical fields, and the design, production, and application of multifunctional materials.







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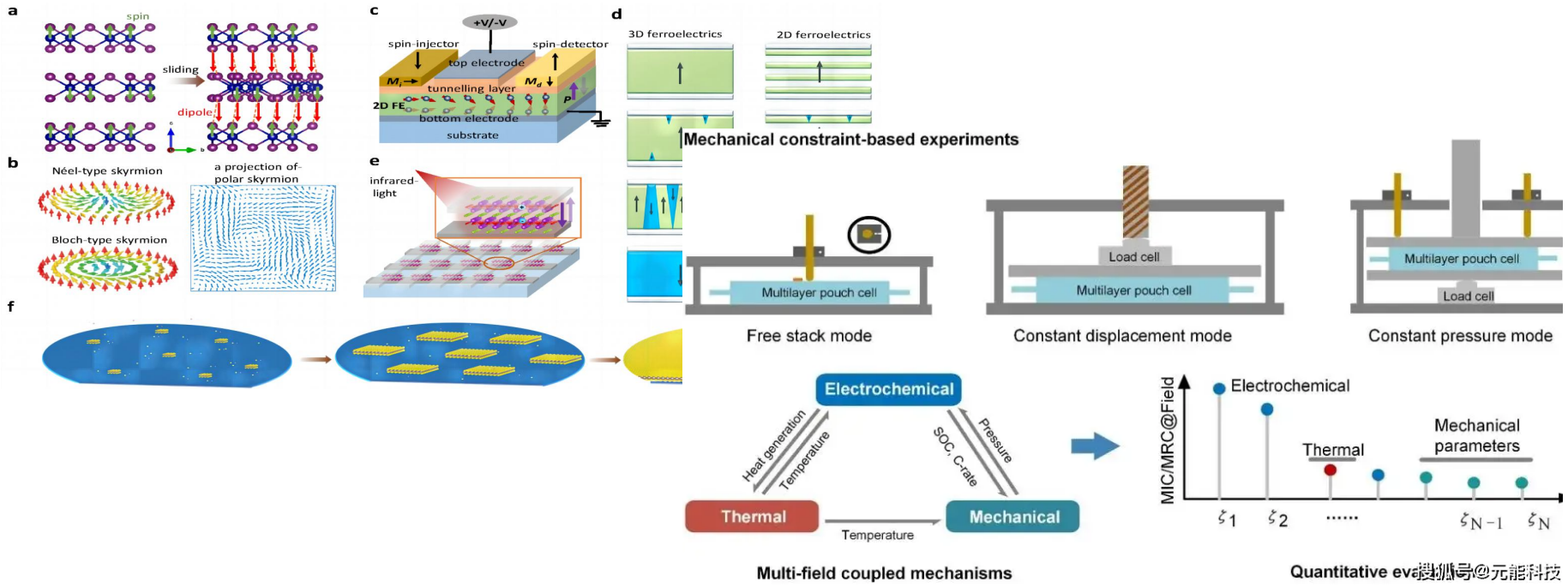
(3) The boundaries between branch disciplines are blurred

Soft matter, gel - non fluid and non solid: fluid - elastic - rigid system dynamics and control



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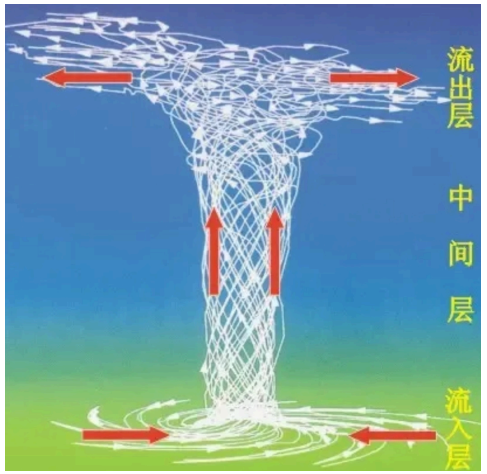
(4) Mechanical problems in extreme environments.

- Ultra high temperature/high pressure/high speed/explosion/ strong impact, microgravity, rarefied gas
- Disaster environment (sandstorms/mudslides/landslides/ landslides/typhoons/tsunamis/corrosion)



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- The application of mechanics in the field of engineering

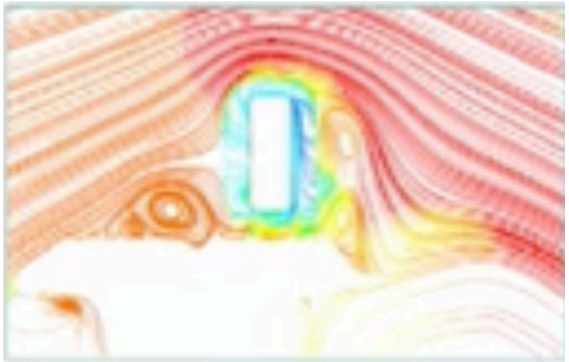
Example: Application of Mechanics in the Field of Civil Engineering

The core role of mechanics in the analysis of disaster mechanisms in civil engineering: Case study - Tacoma Bridge (1940)

analysis of Aviation Engineers: Wind Induced Buckling and Divergent Vibration (Engine Vibration)

Feng Karman et al. in fluid mechanics: Vortex induced resonance (shedding of vortices in blunt H-shaped sections)

Professor Scanlon (1963) from the United States: Separation Flow White Excitation Flutter of Blunt Body Section Bridge Jaw Vibration Theory - Bridge Wind Vibration Theory (Professor Wenport from Canada)



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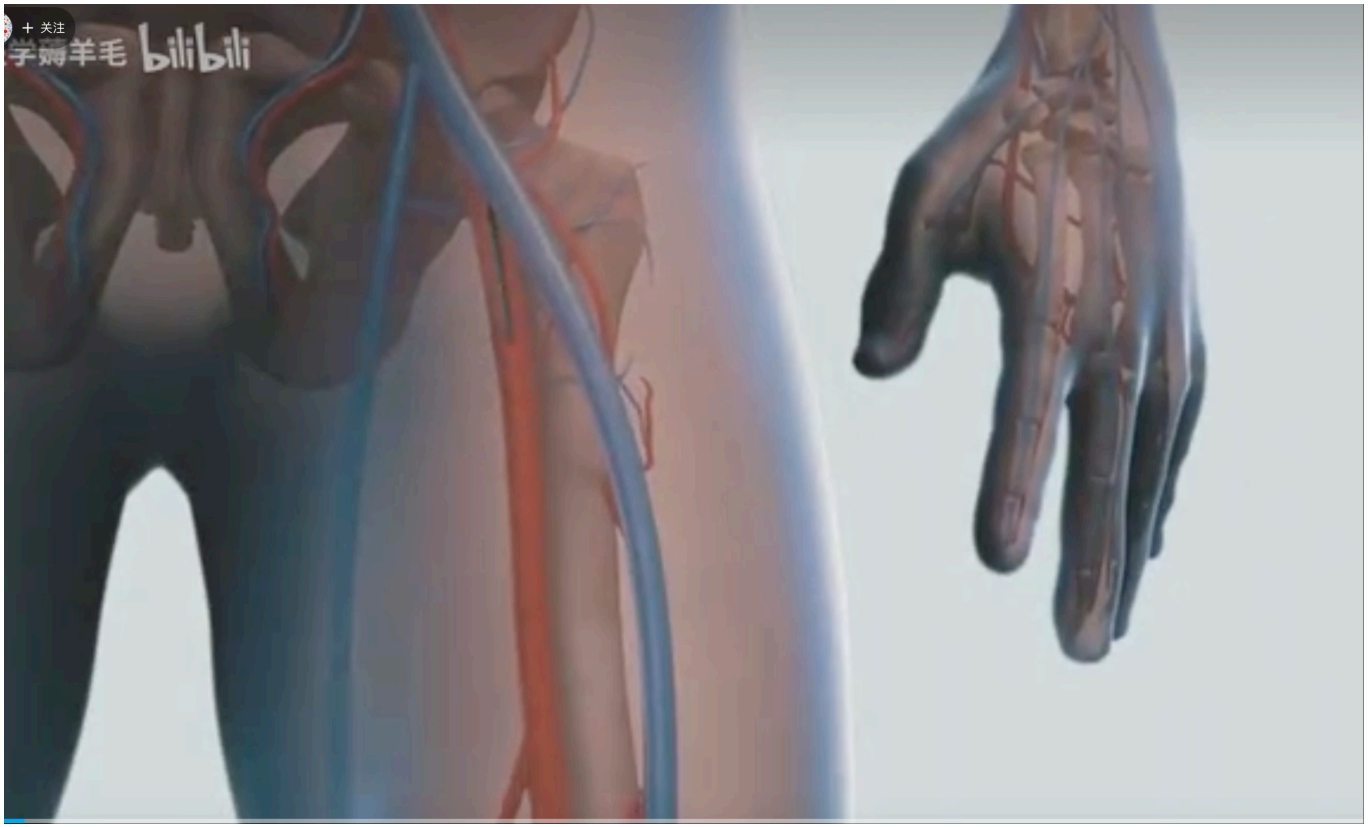
- The Application of Mechanics in the Field of Life

Example： remove thrombus （interventional procedure）



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§0. Introduction

Definition of theoretical mechanics

The science of studying the general laws of mechanical motion of objects.

Mechanical motion refers to the temporal changes in the position and shape of an object in space.

That is, the change in relative "position" between objects or between different parts of the same object over time is called mechanical motion.

And other more complex forms of motion, such as thermal motion, chemical motion, electromagnetic motion, and life phenomena, all involve changes in position, but they cannot be simply attributed to mechanical motion.

The research scope of theoretical mechanics

The mechanical motion of macroscopic objects with speeds much lower than the speed of light.

§0. Introduction

Theoretical Mechanics Research Model

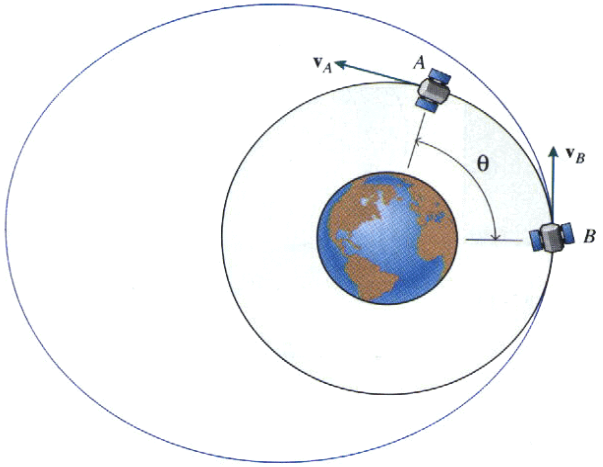
（1） Object Model - Particle and Particle System

particle Discrete particle system Continuum system

（2） Working State Model - Rigid and Deformable Bodies

§0. Introduction

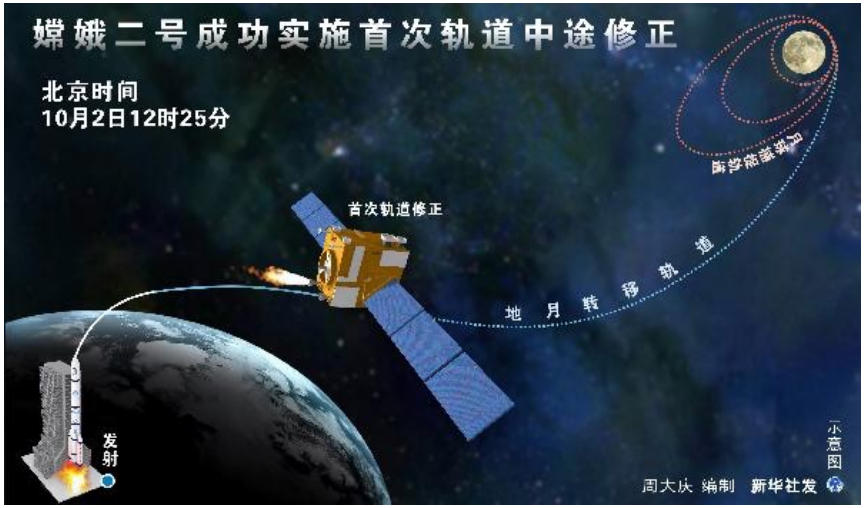
Theoretical Mechanics Research Content



When studying the problem of satellite motion orbit, satellites can be regarded as particles;
But when studying satellite posture control problems, the satellite as a whole is considered a rigid body.

— The size of solar panels installed on satellites is increasing, and their elastic vibrations directly affect the attitude of the main body.

When consider the strength problems of the satellite, it is considered as a deformation body.



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§0. Introduction

- **The research content of statics**
  - **Axioms of Statics**
  - **Constraints and constraint force**
  - **Force analysis and force diagram of objects**
  - **Mechanical models and mechanical diagrams**

§0. Introduction

- Q1: From the development history of mechanics, what roles did experimental observation and mathematics play in establishing the theoretical framework of mechanics?

- Q2: From the perspective of research objects and content, what are the connections and differences between theoretical mechanics and mechanics in college physics?

- Q3: Please provide examples of applying mechanics knowledge from engineering and daily life.

Chapter 1. Fundamentals of statics

- **Axioms of Statics**
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Chapter 1. Fundamentals of statics

Definition：

Static mechanics is the science of studying the equilibrium laws of objects under the action of a force system.

Research object:

The research object of statics is rigid bodies, more strictly speaking, stationary rigid bodies. So, the statics we learn is rigid body statics.

- **rigid body：An object that does not deform when subjected to force.**

An object whose distance between any two points inside remains unchanged under the action of force.

Chapter 1. Fundamentals of statics

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But we know that any object under force (regardless of its magnitude) will undergo deformation.

For example, the lathe spindle undergoes bending deformation during the cutting process;

The crankshaft and connecting rod of an internal combustion engine will undergo bending deformation during motion;

When a vehicle passes over a bridge, the bridge undergoes bending deformation and the piers undergo compression deformation.

The 'rigid body' actually does not exist, it is an idealized and abstract mechanical model.

- **Deformable body: an object that undergoes deformation under the action of force.**

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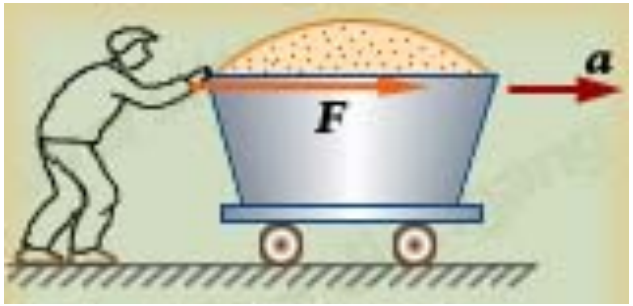
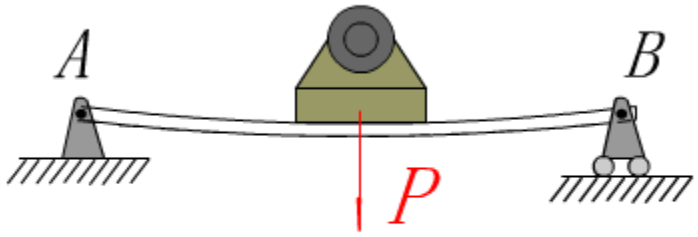
Chapter 1. Fundamentals of statics

- The two elements of statics: rigid body and equilibrium of force system
- **Force**: **The mechanical interaction between objects, which results in a change in the mechanical motion state or shape and size of the objects. (External and Internal Effects)**

External effects



Internal effects



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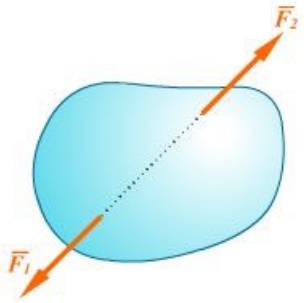
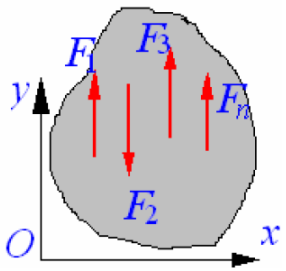
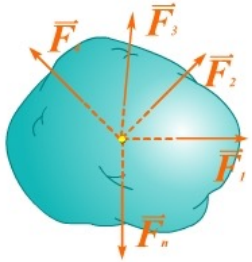
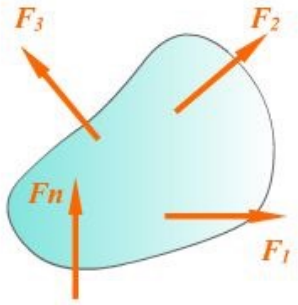
Chapter 1. Fundamentals of statics

- **Force system**：**a group of forces acting on an object**

Plane force system: the lines of action of each force are in the same plane

Space force system: the lines of action of each force are in different planes

Collinear force system	Parallel force system	concurrent forces system	arbitrary force system
			

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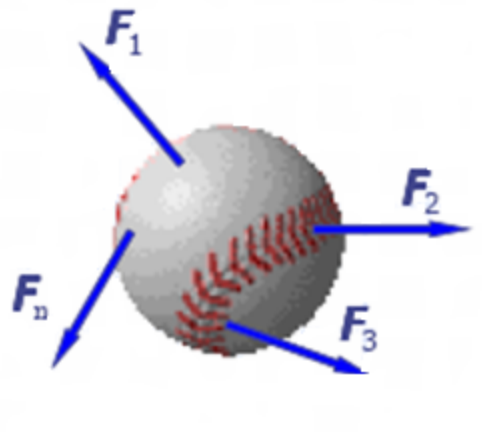
Chapter 1. Fundamentals of statics

- **Force system:** a group of forces acting on an object



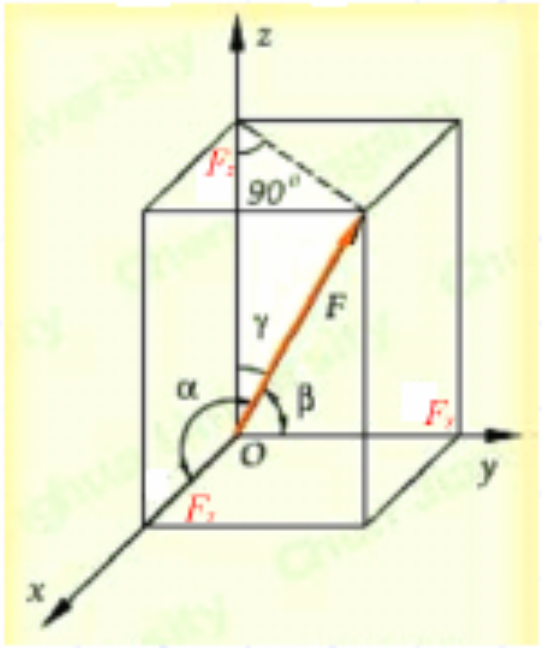
- **Resultant force and component force**

If a force system acting on an object is replaced by a force with the same effect, then that force is called the **resultant** force of the force system, and the forces in the force system are called the **components** of the resultant force.

Chapter 1. Fundamentals of statics

The representation method of force

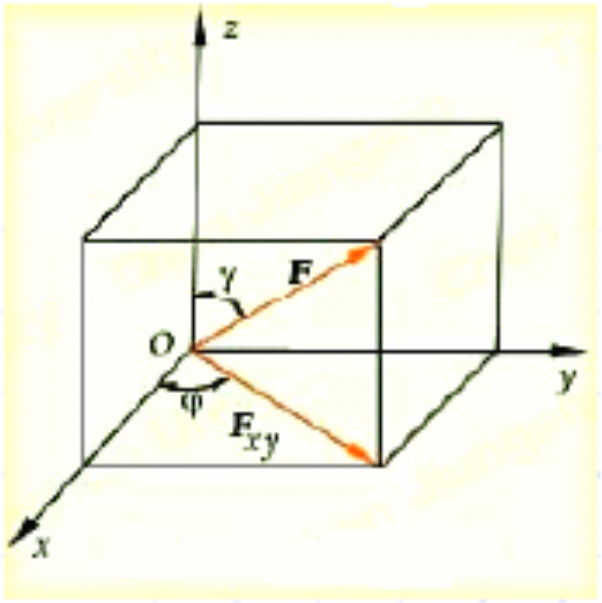
(1) Direct projection method



$$\begin{cases} F_x = F\cos\alpha \\ F_y = F\cos\beta \\ F_z = F\cos\gamma \end{cases}$$

Chapter 1. Fundamentals of statics

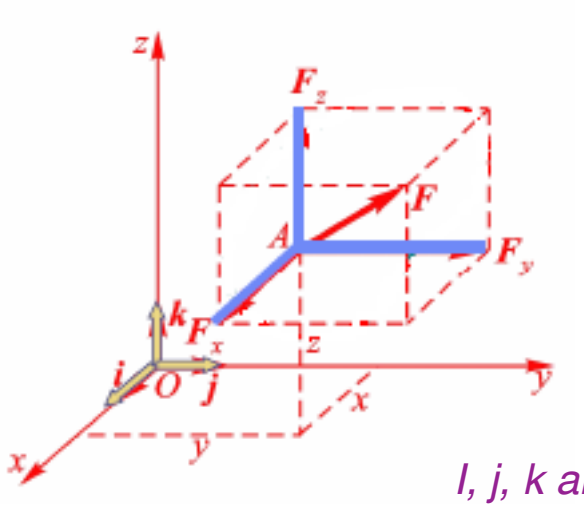
(2) Secondary projection method



$$\begin{cases} F_x = F_{xy}\cos\varphi = F\sin\gamma\cos\varphi \\ F_y = F_{xy}\sin\varphi = F\sin\gamma\sin\varphi \\ F_z = F\cos\gamma \end{cases}$$

Chapter 1. Fundamentals of statics

(3) Analysis expression of force



F_x、F_y、F_z are the projections of vector F on the x、y、z axes.

$$F = F_x i + F_y j + F_z k$$

I, j, k are the unit vectors of the coordinate axes x, y, z respectively.

Chapter 1. Fundamentals of statics

- **Equilibrium ：**

  Objects remain stationary or move at a constant speed in a straight line relative to an inertial reference frame

Equilibrium is a special state of mechanical motion.

Chapter 1. Fundamentals of statics

As for whether an object can be regarded as a **rigid** body in practical problems, it depends not only on the size of the relative **deformation**, but also on the **requirements** of the specific problem.

For example,

(1) when we study the **overall** motion of an airplane, we can consider it as a **rigid** body, and when we study the **strength or stiffness** of wings or certain components, we should consider them as **deformable** bodies

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(2) When studying the overall **equilibrium** problem of cranes, the deformation of each component of the crane is ignored and can be abstracted as a **rigid** body.

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(3) The **pole** used by pole vaulters **cannot** be considered a '**rigid pole**'.

The "pole" in the picture is a large **deformable** body! The deformation that reaches the equilibrium position is related to the force, and the balance of force, geometric description of deformation, and the relationship between force and deformation must be considered simultaneously.

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Small deformation： **linear problem - simple! Easy to solve!**

Large deformation: **nonlinear problem - complex! Generally, numerical methods are used to obtain approximate solutions**

Theoretical mechanics mainly studies the balance of rigid bodies, and the problem of deformation to be studied in subsequent courses such as material mechanics, elastic mechanics, etc. Therefore, the statics content of theoretical mechanics is also known as **rigid body statics**.

Chapter 1. Fundamentals of statics

To solve the problems in statics research, it is necessary to learn the following:

- **Axioms of Statics**
- **Constraints and constraint force**
- **Force analysis and force diagram of objects**
- **Mechanical models and mechanical diagrams**

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- **Axioms of Statics**
- **Constraints and constraint force**
- **Force analysis and force diagram of objects**
- **Mechanical models and mechanical diagrams**

Chapter 1. Fundamentals of statics -
Axioms of Statics

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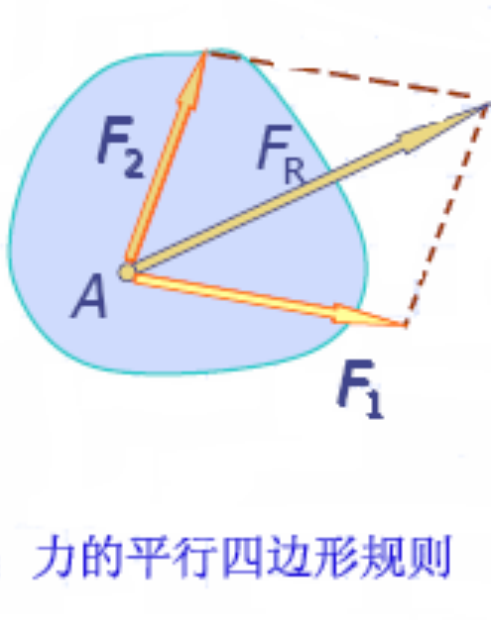
- **Axiom of parallelogram of force**

Two forces acting on the same point on an object can be combined into a resultant force. The point of application of the resultant force is also at this point, and the magnitude and direction of the resultant force are determined by the diagonal of the parallelogram formed by these two forces as edges.



$$F_R = F_1 + F_2$$

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Axioms of Statics

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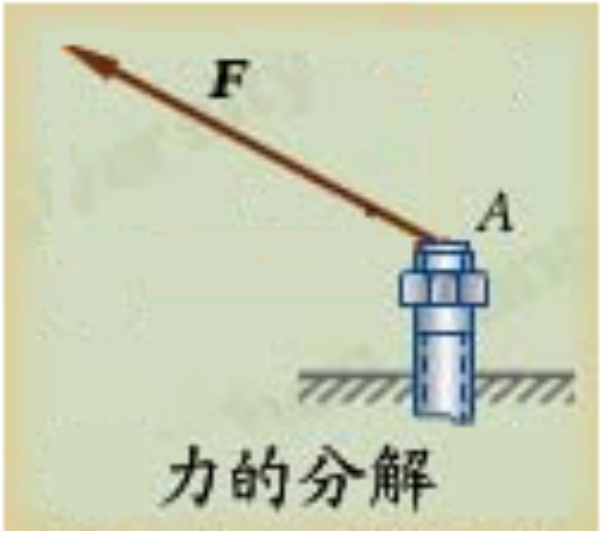
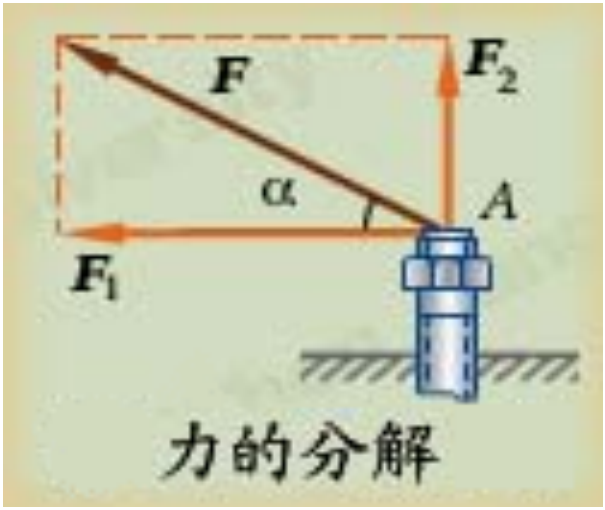
- This axiom provides the basic method for simplifying force systems.
- The parallelogram rule is the composition rule and decomposition rule of forces





Chapter 1. Fundamentals of statics -
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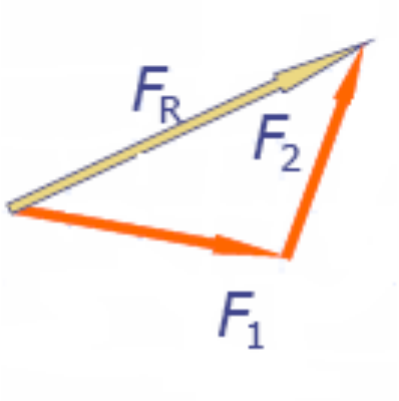
- **Corollary** **The triangle rule of force**



Two forces are connected in sequence, with the resultant force pointing from the beginning of the first force to the end of the second force.

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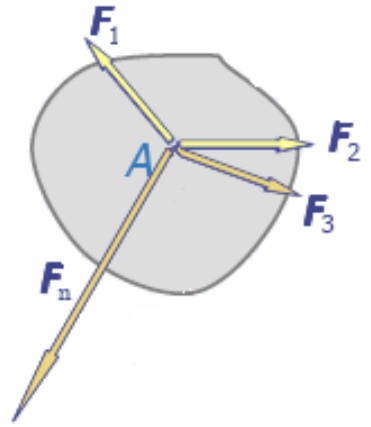
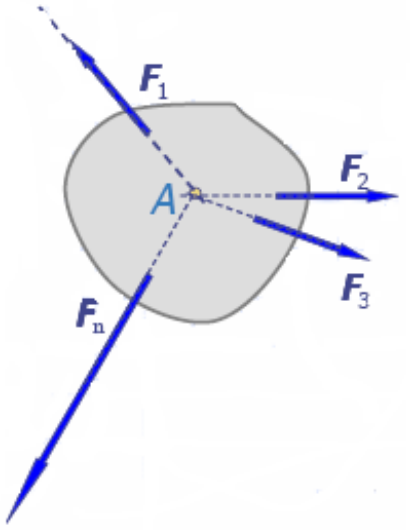
- **Common point force system:** The various forces in a force system have a common point of action.



- **Concurrent force system:** A force system in which the lines of action of each force concurrent at a point is called a convergence force system.



The common point force system is a special type of concurrent force system.

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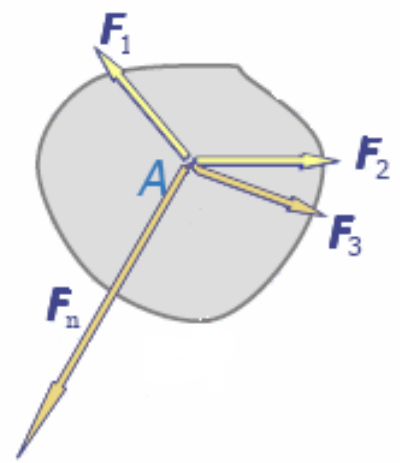
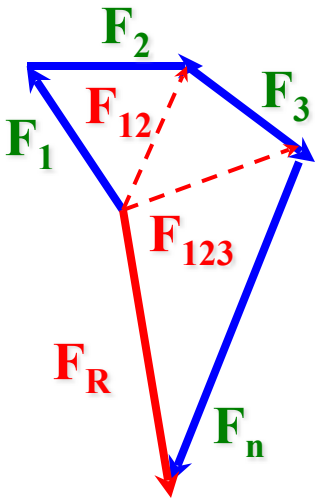
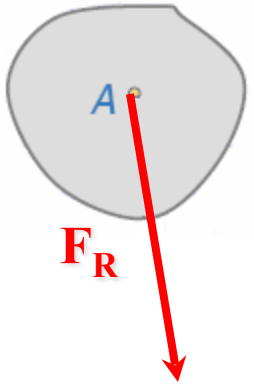
- **Composition of a common point force system**







Polygonal method of force

The magnitude and direction of the resultant force of a common point force system can be represented by the closed edge vectors of the force **polygons** formed by the force vectors in the force system, which act on the **common** point.

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- **Composition of a common point force system**

Analytical expressions for the forces F_i and resultant force F_R in a Cartesian coordinate of a common point force system

$$\mathbf{F}_i = F_{xi}\mathbf{i} + F_{yi}\mathbf{j} + F_{zi}\mathbf{k} \quad , \quad \mathbf{F}_\mathrm{R} = F_{\mathrm{R}x}\mathbf{i} + F_{\mathrm{R}y}\mathbf{j} + F_{\mathrm{R}z}\mathbf{k}$$

According to the projection theorem of resultant forces

$$F_{\mathrm{R}x} = \sum_{i=1}^{n} F_{xi}, \quad F_{\mathrm{R}y} = \sum_{i=1}^{n} F_{yi}, \quad F_{\mathrm{R}z} = \sum_{i=1}^{n} F_{zi}$$

Furthermore, obtain the magnitude and direction cosine of the resultant force of the common point force system

$$\left.\begin{aligned} F_\mathrm{R} &= \sqrt{F_{\mathrm{R}x}^2 + F_{\mathrm{R}y}^2 + F_{\mathrm{R}z}^2} \\ \cos(\boldsymbol{F}_\mathrm{R}, \boldsymbol{i}) &= \frac{F_{\mathrm{R}x}}{F_\mathrm{R}}, \quad \cos(\boldsymbol{F}_\mathrm{R}, \boldsymbol{j}) = \frac{F_{\mathrm{R}y}}{F_\mathrm{R}}, \quad \cos(\boldsymbol{F}_\mathrm{R}, \boldsymbol{k}) = \frac{F_{\mathrm{R}z}}{F_\mathrm{R}} \end{aligned}\right\}$$

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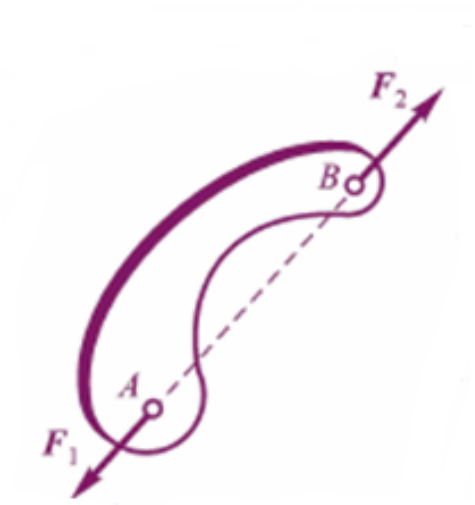
Axioms of Statics

- **Axiom of Two Forces Equilibrium**

The sufficient and necessary condition for the equilibrium of a rigid body subjected to two forces is that these two forces are equal in magnitude, opposite in direction, and along the same line of action.



$$F_1 = F_2$$

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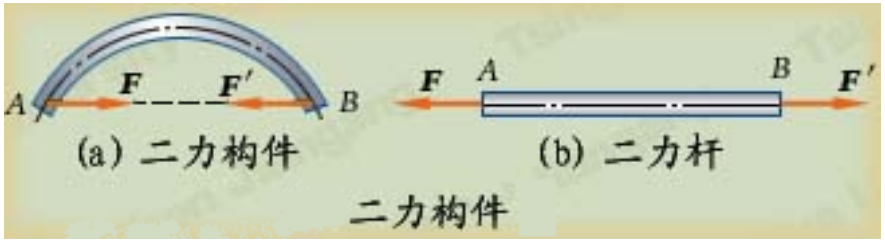
- **Axiom of Two Forces Equilibrium**

This axiom applies to '**rigid bodies**'. This axiom indicates the conditions that must be met when the simplest force system acting on a rigid body is in equilibrium. It is the foundation for handling complex force system equilibrium.

A rigid body that only keeps equilibrium under the action of two forces is called a two force body or a two force component.

When the component is a straight rod, it is called a two force rod.



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- **Axiom of adding removing equilibrium force system**

Adding or removing any equilibrium force system from a known force system does not change the effect of the force system on the rigid body.

The new force system obtained by adding or romoving the equilibrium force system is equivalent to the original force system.

This axiom is an important basis for studying the equivalence of force systems.

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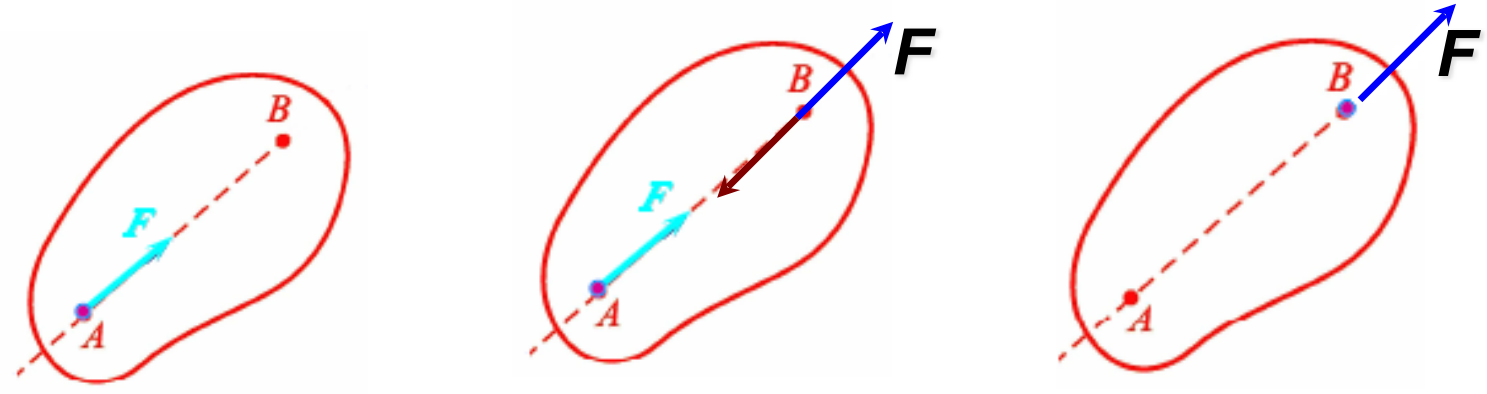
- **Corollary 1: Transmissibility of Forces**

The force acting on a rigid body at a certain point can move along its line of action to any point inside the body without changing the effect of the force on the rigid body.



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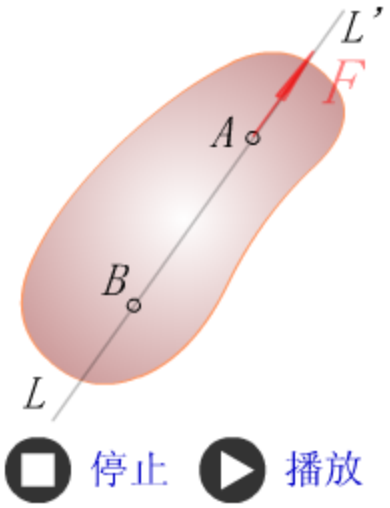

The force acting on a rigid body is a "sliding vector", which means that the force vector can move along the line of action to any point.

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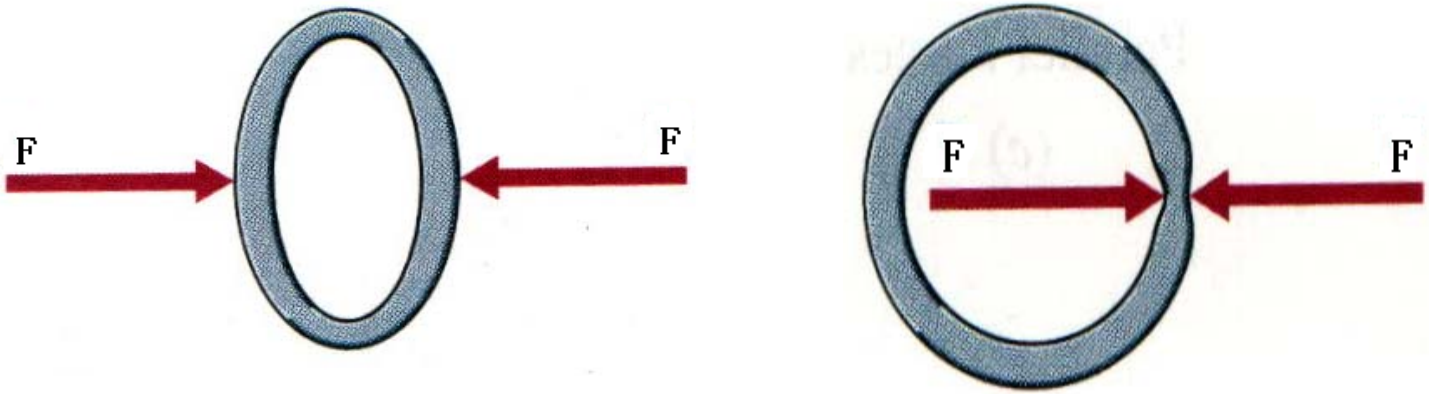

The principle of force transmissibility only applies to rigid bodies and not to deformable bodies.

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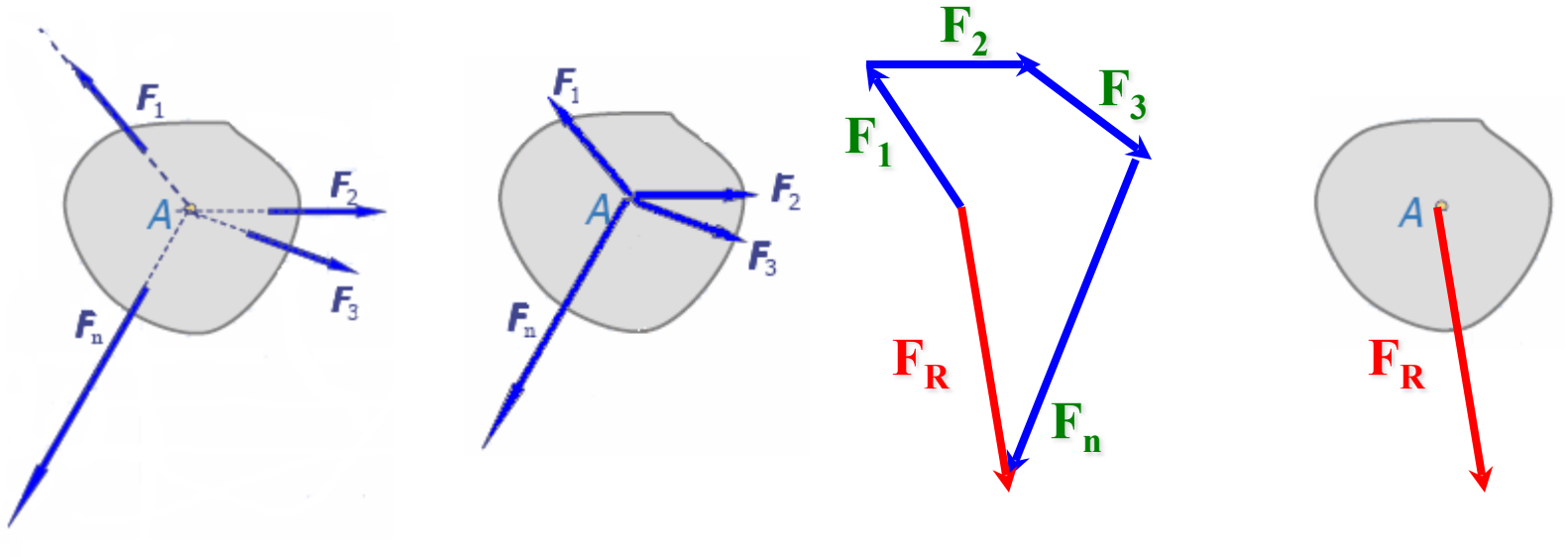
- **composition of concurrent forces**



The magnitude and direction of the resultant force in the concurrent force system can be represented by the closed edge vectors of the force polygons formed by the force vectors in the force system; Its action lines passes through the concurrent point (**action line**).

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- **composition of concurrent forces**

Analytical expressions for the forces F_i and the resultant force F_R of the concurrent force system in a Cartesian coordinate system

$$\mathbf{F}_i = F_{xi}\mathbf{i} + F_{yi}\mathbf{j} + F_{zi}\mathbf{k} \quad , \quad \mathbf{F}_R = F_{Rx}\mathbf{i} + F_{Ry}\mathbf{j} + F_{Rz}\mathbf{k}$$

According to the projection theorem of resultant forces

$$F_{Rx} = \sum_{i=1}^{n} F_{xi}, \quad F_{Ry} = \sum_{i=1}^{n} F_{yi}, \quad F_{Rz} = \sum_{i=1}^{n} F_{zi}$$

obtain the magnitude and direction cosine of the resultant force of the concurrent force system

$$\left.\begin{aligned} F_R &= \sqrt{F_{Rx}^2 + F_{Ry}^2 + F_{Rz}^2} \\ \cos(\boldsymbol{F}_R, \boldsymbol{i}) &= \frac{F_{Rx}}{F_R}, \quad \cos(\boldsymbol{F}_R, \boldsymbol{j}) = \frac{F_{Ry}}{F_R}, \quad \cos(\boldsymbol{F}_R, \boldsymbol{k}) = \frac{F_{Rz}}{F_R} \end{aligned}\right\}$$

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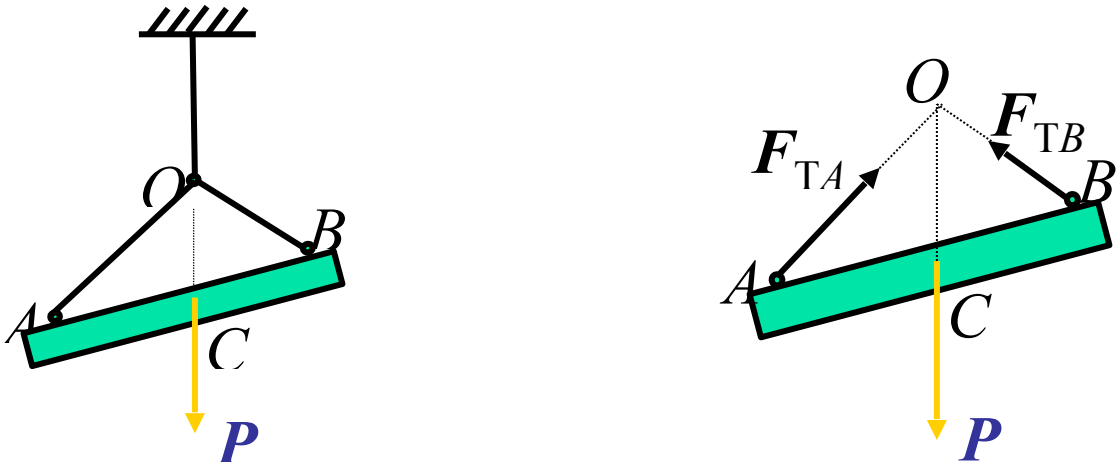
Corollary 2 The Three Concurrent Forces Theorem

The line of action of the third force passes through the point of intersection, if three mutually balanced forces act on a rigid body and the lines of action of two of them intersect at a point.



Note: When two of the three forces intersect, the concurrent of the three forces is a **necessary** condition for the balance of the three forces, **not** a **sufficient** condition, because any force system in which three forces act on the same plane and intersect at a point is obviously not necessarily balanced.

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- **Action and Reaction Principle**

Whenever one object exerts a force on another object, the second object exerts an equal and opposite on the first.

This axiom summarizes the relationship of interactions between objects, indicating that forces and reactions appear in pairs and act on different objects, each producing its own effect For contact forces, the third law is always correct, but for non-contact forces, this law may not be correct.

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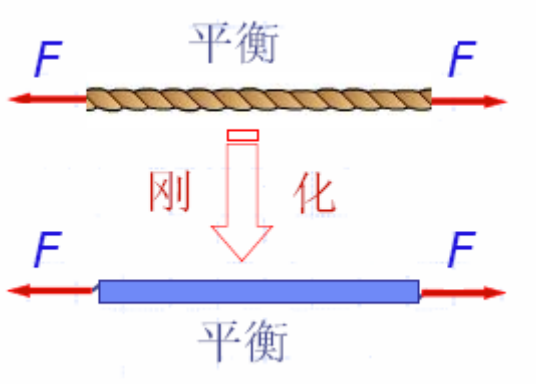
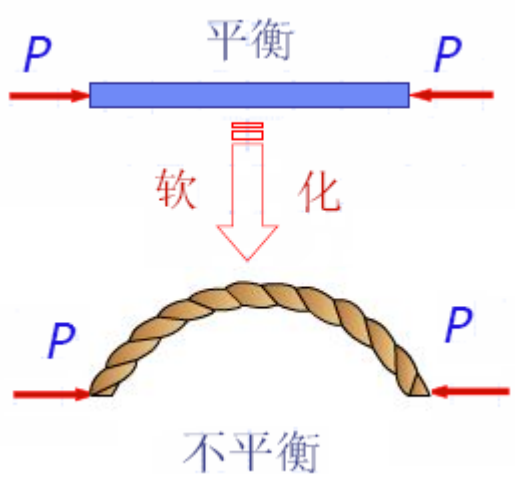
- **Axiom of rigidization**

The deformable body is in equilibrium under the action of a known force system, and remains in equilibrium even after stiffening.





This axiom provides the condition for treating deformable bodies as rigid bodies.
The equilibrium condition of a rigid body is a necessary condition for the equilibrium of a deformable body, rather than a sufficient condition.

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Question 1: Is the following statement correct?

A）An object in equilibrium can be considered a rigid body.

B）Objects with small deformations can be regarded as rigid bodies.

C）When studying the mechanical motion of an object, if the deformation of the object has no or minimal impact on the problem being studied, the object can be considered a rigid body.

Question 2: Is the following statement correct?

A）Force is a sliding vector that can move along the line of action.

B）The effects of force on objects can be divided into external effects (motion effects) and internal effects (deformation effects), and the main focus of theoretical mechanics is on the external effects of force.

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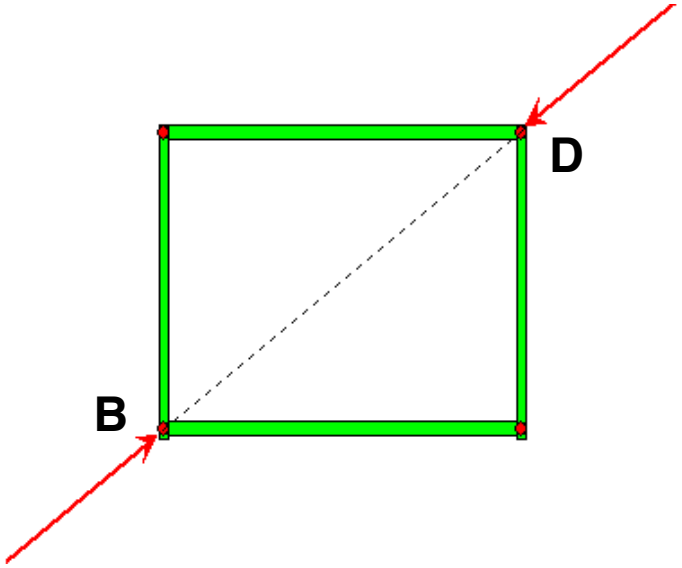
Questions

Question 3: Four weightless members are hinged as shown in the figure. Can this system be balanced by adding a pair of equal, opposite, and collinear forces to two points on the BD? Why?



Question 4: What is the scope of applicability of the transmissibility of force?

Question 5: What is the scope of application of the parallelogram axiom of force and the axioms of action and reaction forces?

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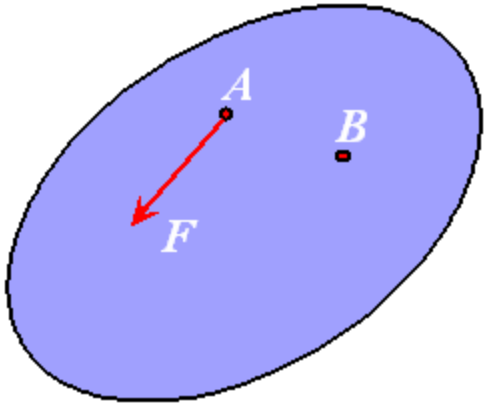
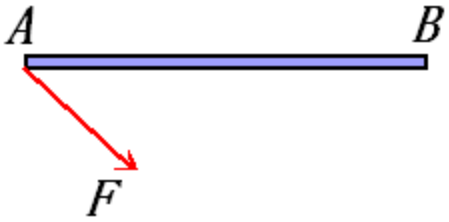
Questions

Question 6: A point on a rigid body is subjected to a force *F*, as shown in the figure. Can a force be added to point B to make the rigid body balanced? Why?





Question 7: Could vector expression of known force F, $F = F_x i + F_y j + F_z k$, fully express the three elements of the force? Why?

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Questions

Question 8: If an object is balanced under the action of a certain force system, according to the Axiom of Addition and Removing of Balance Force System, and an equilibrium force system is added to this object, is the object still in equilibrium? Why?

Question 9: Regarding the rigid body shown in the figure, based on the transferability of force, can force F be moved from point A on the rigid body to a point outside the rigid body